

1,500 Greek passages · 61 expert-identified names · 1,781 untested leads

pharmakon

2,000 years of Greek plant medicine, measured.

An uncertainty-aware map from Dioscorides' 1st-century Greek to modern species, chemistry and testable research leads.

EthnoHack 2026 · Track 1 — Traditional knowledge

pharmakon.adoranto737.workers.dev · github.com/LSUDOKO/Pharmakon

Dioscorides described about 600 medicinal plants. Which ones did he mean — and was he right?

Names don't map to species

An ancient name can point at several modern plants, and translations hide that doubt. One wrong identification breaks every lead built on it.

Survival is studied by hand

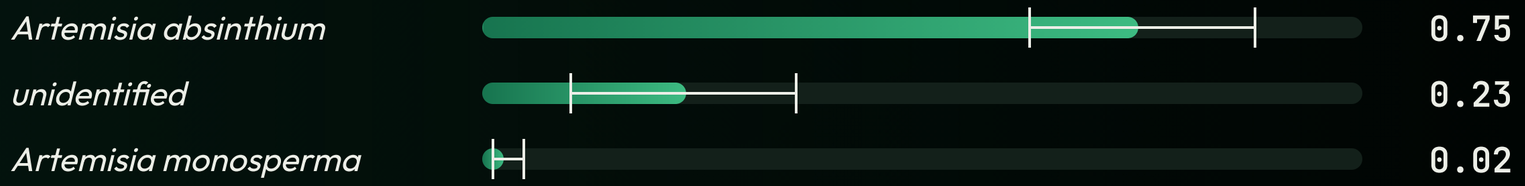
Whether a remedy persisted from Dioscorides through Byzantium to today's villages has only been traced plant by plant.

Predictions are rarely tested

Herb–disease models are almost never checked with a time split, so "history predicts science" is assumed, not measured.

ἀψίνθιον

De materia medica 3.23



Pharmakon's answer for wormwood, with 90% bootstrap intervals — doubt kept, not hidden.

Data: one text chain, eight open databases

1st c.

Dioscorides

1,500 passages, **835** chapters
Wellmann ed., First1KGreek TEI

2nd c.

Galen

476 drug entries, books 6–8
Kühn ed., First1KGreek TEI

13th c.

John the Physician

61 expert-identified names
Lardos et al. 2024 (CC BY)

today

Greek villages

1,226 modern use records
Pelion 2023 · N. Aegean 2018

GBIF
taxonomy, Greek occurrences

Wikidata · LOTUS
10,252 plant compounds

ChEMBL 37
455 human targets

Open Targets
target → disease

NLM MeSH
90 verified conditions

PubMed
1,999 plant–condition pairs labelled

PubChem
toxicity citations (GHS)

LSJ · Wikipedia
lexicon, species descriptions

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Vienna Dioscorides (Codex Vindobonensis med. gr. 1, 6th c.), public domain — the tradition this project measures.

Method I — read the Greek, identify the plant

Every use keeps its Greek words

1. **Rule lexicon** — Greek condition stems, clause by clause, exact spans.
2. **LLM reading** — 1,158 passages, 5,075 claims; finds 95% of rule uses.
3. **Verify** — every span must be a substring of the source: **100%** pass. MeSH only via NLM lookup.

2,561 → 3,550

usable claims after the verified LLM layer; the backtest keeps the original 2,561

P(species | name), with an "unidentified" option

- geography — GBIF records in Greece, Cyprus, Turkey
- name continuity — modern Greek names, Latinised epithets
- scholarly agreement — LSJ, botanical historians
- genus consensus
- plant-description match — 324 Greek chapters vs 350 species

Conditional logit → temperature scaling → bootstrap intervals. No pharmacology inside: identity is decided before any drug evidence is seen.

“δύναμιν δὲ ἔχει στυπτικήν, θερμαντικήν, ἀποκαθαρτικὴν τῶν ἐνηρεικότων στομάχῳ καὶ κοιλίᾳ χολωδῶν.”

→ **stomach complaints**

use group GI · part whole · verified span, CTS 3.23.1

Method II — trace each use, then test with a time machine

Survival across 2,000 years

Dioscorides → Galen → John the Physician → modern Greece

For each name × body system: is the use attested in each era? Compared with a **label-shuffle null** (1,000 permutations), so chance overlap is not mistaken for tradition. Galen is checked for copying (word 4-gram containment).

Rolling-origin backtest

Stand at 2000, 2005, 2010, 2015. Rank plant–condition pairs nobody had studied, using only what was known then. Check PubMed up to 2025.

- ✓ features dated ≤ cutoff (compound, ChEMBL and study years)
- ✓ research-popularity baseline to beat
- ✓ no LLM output, no LLM-derived claims, frozen non-LLM identification
- ✓ cluster-bootstrap 95% intervals at every cutoff

2000

1,404 untested pairs
122 studied by 2025

2005

1,384 untested pairs
102 studied by 2025

2010

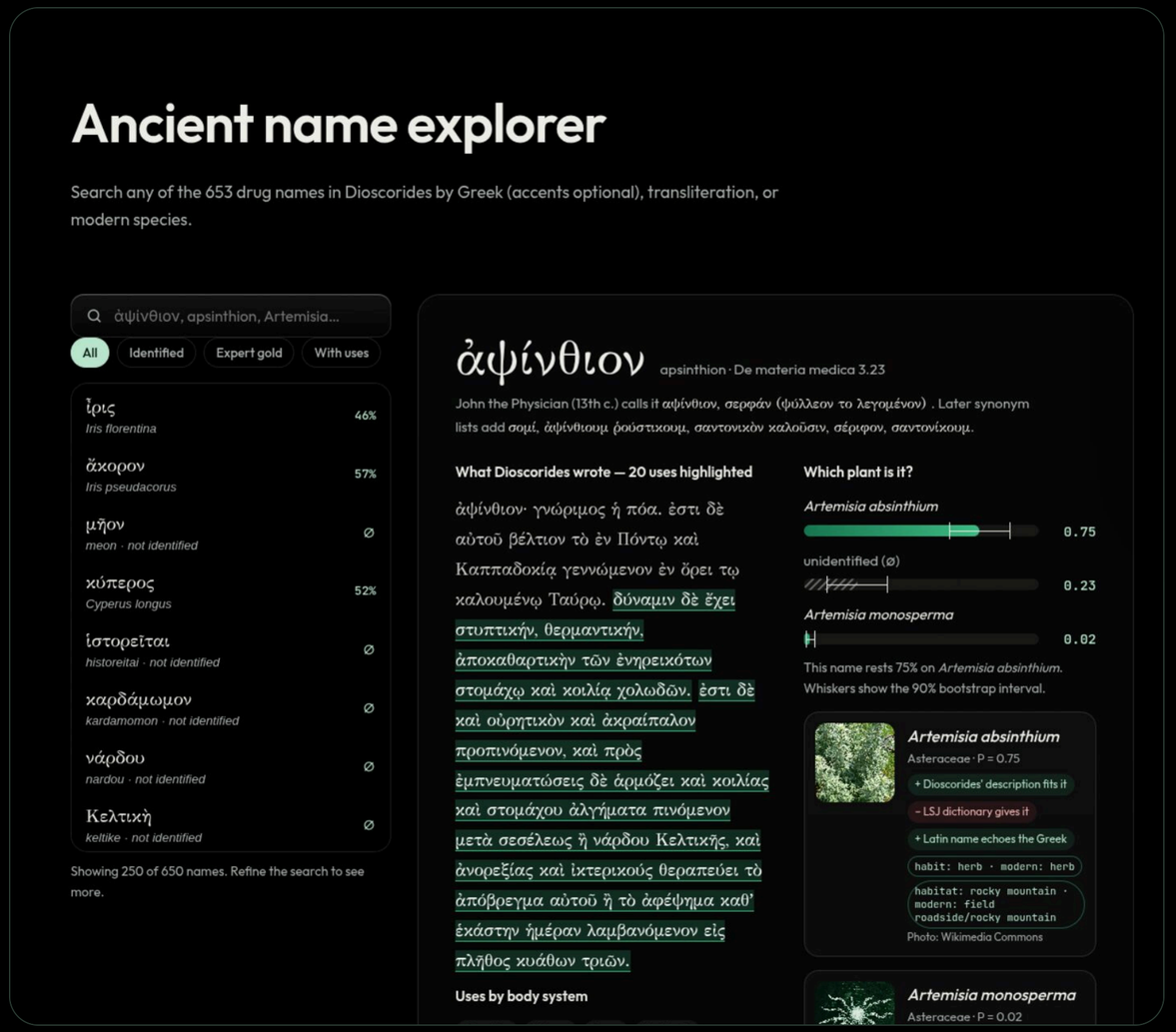
1,361 untested pairs
79 studied by 2025

2015

1,332 untested pairs
50 studied by 2025

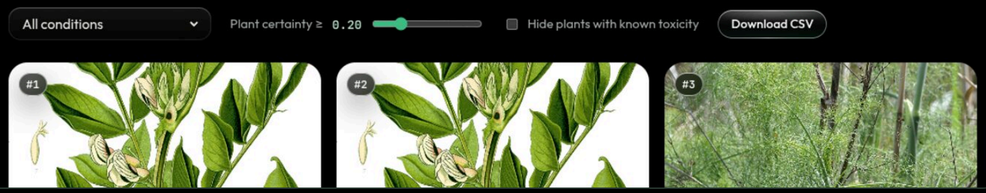
All body systems. The skin/wound pilot is kept for comparison.

Prototype — live, open, usable from code and AI assistants



Leads for the lab

Plant–condition pairs with no published pharmacology yet, ranked by forecast score × how sure we are of the plant. Hypotheses to test, never advice to use.



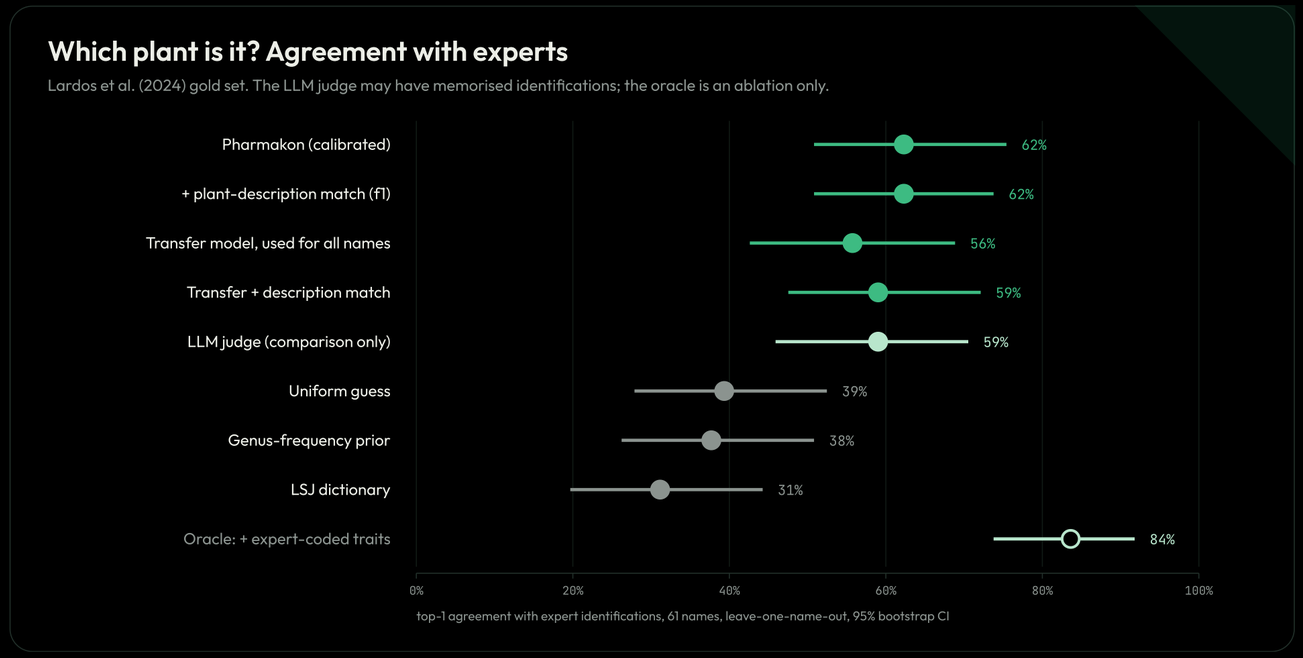
Website · JSON API · MCP server

Explorer with highlighted Greek, candidate photos, probability bars, description matches and a four-era survival table. Leads with toxicity flags, evidence paths and the exact PubMed query.

```
claude mcp add --transport http pharmakon \
  https://pharmakon.adoranto737.workers.dev/mcp
```

6 read-only tools · CSV export · Streamlit research dashboard · 27 tests · one-command reproduction

Result I — identification beats the dictionary



62%

agreement with expert identifications, vs **31%** for the LSJ dictionary

0.50

Brier score, vs 1.34 for the dictionary (lower is better)

0.21 → 0.14

calibration error once Dioscorides' plant descriptions are compared with modern ones. Target ≤ 0.10: **not yet met**.

Leave-one-name-out over 61 names, 95% bootstrap CIs. The LLM judge (59%) is a comparison only — it may have memorised published identifications. Expert judgement is not ground truth.

Result II — the uses survived; history did not predict science

Did the uses survive 2,000 years?

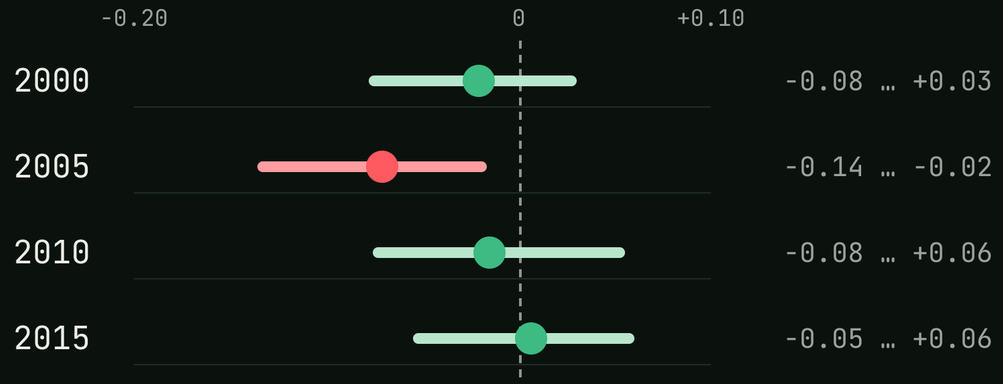
Dioscorides uses whose plant appears in the modern Greek record (508 name × use rows)

Observed	48.6%
Shuffled labels, average	40.8%
Shuffled labels, 95th percentile	43.3%

p = 0.001 over 1,000 label shuffles. Galen re-attests **37%** of the uses, John the Physician **34%**. Caveat: one modern study covers skin uses only.

Did history predict what science studied next?

Δ AUPRC = (history + popularity) − popularity, 95% CI; left of zero = history hurts



An honest null. Research popularity predicts which pairs get studied; 2,000 years of history never adds to it, and at 2005 it hurts. Going from the skin pilot to all body systems (132 → 353 positives) made the null sharper, not weaker.

Impact — 1,781 uncertainty-weighted leads for the lab



#1 *Vicia faba*

Eye Diseases · P = 0.66 · path ✓



#3 *Foeniculum vulgare*

Kidney Diseases · P = 0.58



#4 *Artemisia absinthium*

Eye Diseases · P = 0.75 · path ✓



#5 *Cupressus sempervirens*

Wounds and Injuries · P = 0.66 · path ✓



#7 *Ruta graveolens*

Eye Diseases · P = 0.52 · path ✓



#8 *Punica granatum*

Hernia · P = 0.73 · path ✓

16/20

top leads with a closed compound → human target → disease path (LOTUS · ChEMBL · Open Targets)

5/20

top leads that change if plant uncertainty is ignored (Kendall τ 0.50)

2/20

carry a PubChem-cited toxicity flag — shown, never hidden

Who uses it: pharmacognosy labs choosing what to screen next · historians reading Dioscorides with evidence · conservation (IUCN status per species). Every lead keeps its exact PubMed query.

Limits, ethics and what comes next

What we don't claim

- Expert gold is judgement, not ground truth (61 names, wide intervals).
- "Supported" = PubMed co-mention proxy; human audits generated, not yet filled.
- Calibration 0.14 still misses ≤ 0.10.
- LLM outputs are verified word-for-word but not yet expert-audited.

Responsible research

- Research hypotheses only — no dosing, no medical advice.
- Toxic genera flagged with PubChem citations.
- No LLM output ever feeds the backtest (tested).
- Nagoya Protocol: benefit-sharing for any commercial follow-up.

Next

- Expert audit of claims, conditions and labels.
- Richer floras (Flora of Greece, POWO with permission).
- Paul of Aegina, Oribasius; Arabic and Latin transmission.
- Screening partnership for top non-toxic leads; Zenodo DOI.

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